Section A — Concept Application

S 1 SCENARIO : Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

Answer:

1. CPU: Choose a fast processor with multiple cores to run programming software, browsers, and video calls together.
2. RAM: At least 8 GB RAM is suitable for basic multitasking. 16 GB is recommended for better performance.
3. Storage: Choose an SSD because it starts the computer and opens applications faster than an HDD.
4. Compatibility: Check that the CPU, RAM, and storage are compatible with the motherboard.
5. Budget: Select hardware according to the company's needs and budget.

S 2 SCENARIO: After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Answer: Antivirus software protects computers from viruses and malware, but it can slow down the system.

1. CPU: Antivirus scanning uses CPU power, which may slow down other applications.
2. RAM: Antivirus software uses RAM to run its background processes.
3. Storage: Scanning files uses disk resources, which can slow down file access.
4. Solution: Schedule scans during non-working hours and keep the antivirus software updated

S 3 SCENARIO: A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

Answer:

1. Check whether the keyboard and printer are connected properly.
2. Try another USB port or cable.
3. Restart the computer.
4. Open Device Manager and check for driver errors.
5. Update or reinstall the keyboard and printer drivers.
6. Run Windows Update to fix operating system issues.
7. Test the devices again.

S 4 SCENARIO Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations

Answer:

1. UPS (Uninterruptible Power Supply): Provides temporary battery power during a power cut.
2. Data protection: Gives users time to save their work and shut down the computer safely.
3. Surge protector: Protects computer components from sudden voltage spikes.
4. Hardware protection: Helps prevent damage to the computer, monitor, and other connected devices.

S 5 SCENARIO New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

Answer:

DNS (Domain Name System): DNS converts a domain name, such as server.company.local, into an IP address.

DHCP (Dynamic Host Configuration Protocol): DHCP automatically assigns an IP address and other network settings to devices.

How they work together:

1. DHCP provides the device with an IP address, subnet mask, default gateway, and DNS server address.
2. DNS uses name records to find the IP address of the requested server.
3. The device uses that IP address to connect to the server.

Most likely problem:

- If devices can access the server by IP address but not by name, the DNS configuration is likely incorrect.
- If devices cannot obtain a valid IP address, the DHCP service or its configuration may be incorrect.

S 6 SCENARIO After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question: Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

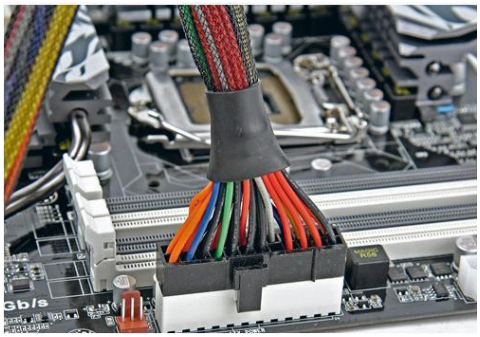
Answer:

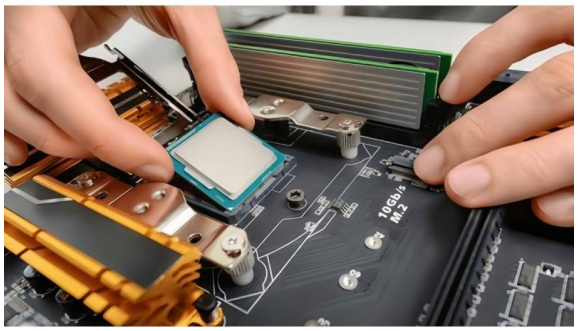
1. Check the application server's IP address and confirm that the server is running.
2. Test connectivity using the `ping` command, if ICMP traffic is allowed.
3. Check the firewall rules for blocked traffic.
4. Verify the source IP address, destination IP address, port number, and protocol.
5. Check the firewall logs to identify denied connections.
6. If a legitimate connection is blocked, create or correct a firewall rule to allow the required traffic.
7. Test the application again and confirm that the connection works.

Section B — Practical Application

Task 1: Workstation Assembly Physically assemble a workstation for the startup office and document the component installation sequence. Install CPU, RAM, storage, and other core components into the chassis in the correct sequence. Connect all required internal cables (power, data, front panel). Document each installation step in order. System must pass POST (Power-On Self-Test) after assembly.

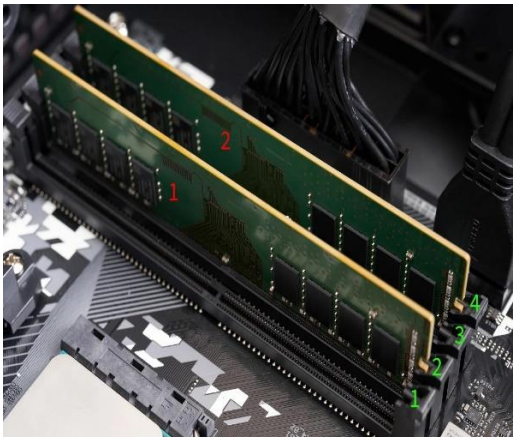
1. Switch off and unplug the computer.
2. Install the CPU on the motherboard.
3. Install the CPU cooler and RAM.
4. Install the SSD or HDD.
5. Fix the motherboard and power supply inside the cabinet.
6. Connect the power cables, storage cables, and front-panel cables.
7. Connect the monitor, keyboard, and power cable.
8. Switch on the computer and check the POST screen and BIOS.



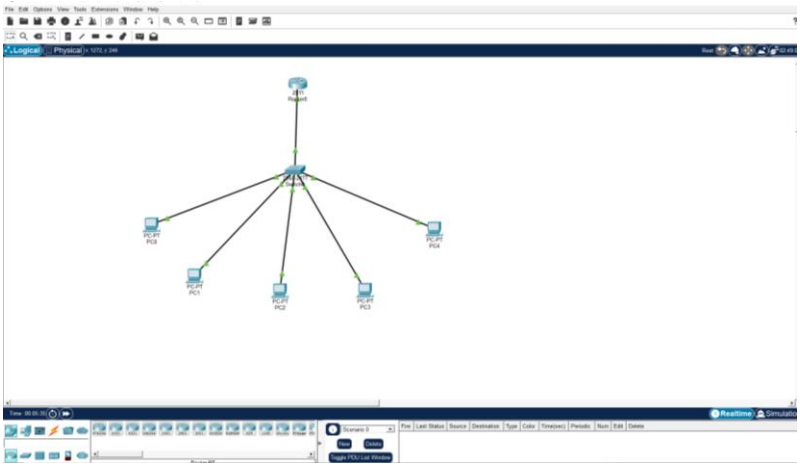


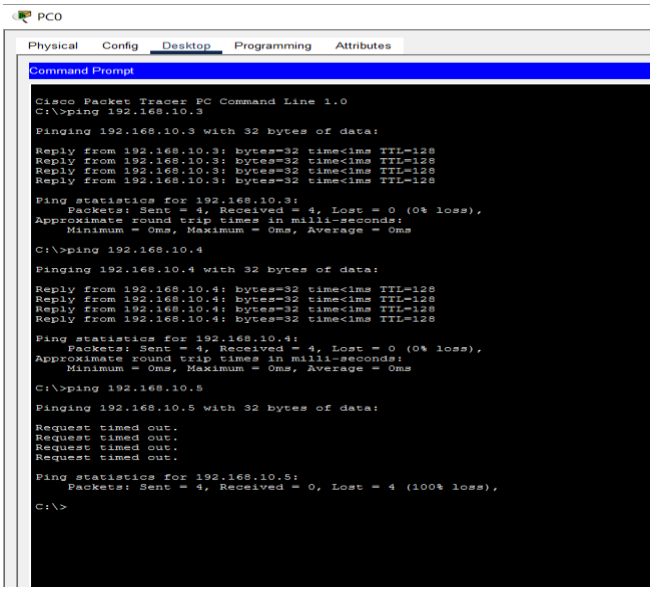
Task 2: RAM Mismatch Troubleshooting Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings on a startup office workstation. Identify the RAM mismatch error via BIOS/UEFI POST messages. Determine the root cause (mismatched speed, timing, or capacity). Resolve the mismatch by reconfiguring or reseating the RAM correctly. Confirm the system boots normally after the fix.

- Start the computer and check for RAM errors on the POST screen.
- Enter BIOS/UEFI by pressing `F2` or `Del`.
- Check the RAM capacity, speed, and settings.
- Switch off and unplug the computer.
- Remove and reseat the RAM correctly.
- Use compatible RAM and supported BIOS settings.
- Start the computer again and check the RAM.
- Confirm that the computer boots normally.



Task 3: LAN Topology Build Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet. Use Cisco Packet Tracer to build the topology. Include 5 end-systems and 1 network printer. Assign IP addresses from a Class C subnet to all devices. All assigned IPs must be reachable via Ping.





Task 4: Clean OS Installation Perform a clean OS installation with manual disk partitioning and startup optimization on a startup office workstation. Perform a clean install of Windows/Linux from ISO. Manually partition the disk (not guided/automatic partitioning). Optimize startup programs/services for faster boot. OS must boot without errors after installation.

- Download a Windows/Linux ISO file.
- Open VirtualBox and create a new virtual machine.
- Select the ISO file and allocate RAM, CPU, and disk space.
- Start the virtual machine and begin OS installation.
- Select manual disk partitioning and create the required partitions.
- Complete the installation and restart the virtual machine.
- Open Task Manager → Startup Apps in Windows.
- Disable unnecessary startup applications.
- Restart the system and check that it works properly.

Section C — Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module

Your project must: Identify the three hardware components most likely to cause random reboots under load. Explain why the systems pass POST in the morning but fail in the afternoon. Describe how you would use a stress-test tool to confirm a thermal or power issue. Propose three specific steps to remediate the frequent system restarts. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Answer:

1. Three hardware components most likely to cause random reboots

1. CPU: It may overheat during heavy workloads.
2. PSU (Power Supply Unit): It may not provide enough power to the components.
3. GPU (Graphics Card): It may overheat or consume too much power during rendering.

2. Why do systems pass POST in the morning but fail in the afternoon?

In the morning, the computers are cool and under low load. During afternoon rendering, CPU and GPU temperatures increase. Poor ventilation inside the desk cabinets traps hot air, which may cause overheating and random restarts.

3. How to use a stress-test tool

1. Check CPU and GPU temperatures using a hardware monitoring tool.
2. Run a CPU/GPU stress test under supervision.
3. Monitor temperatures, power usage, and system stability.
4. Stop the test if temperatures exceed safe limits or the system becomes unstable.

4. Three remediation steps

1. Improve ventilation by moving computers out of closed cabinets and cleaning dust from fans.
2. Check CPU/GPU cooling systems and replace faulty fans or thermal paste if required.
3. Test the PSU and replace it if it cannot provide sufficient, stable power.

5. Why might a PSU upgrade be necessary?

A PSU upgrade is necessary if the existing PSU cannot provide enough stable power during heavy rendering workloads. If the PSU has sufficient capacity and works correctly, upgrading it may not solve the problem. Temperature and power tests should be performed before replacing it.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer.

The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address. The plan must allocate addresses for all 5 systems and 1 printer named in the scenario. Instruct the AI to also identify which addresses to reserve for future expansion. Document the exact prompt(s) used.

Nexus Innovations – IP Addressing and Subnetting Plan

Aim: To create an IP addressing plan for 5 computers and 1 network printer using a private Class C IPv4 subnet.

1. Network Details

Network detail	Value
Network address	192.168.10.0
Subnet mask	255.255.255.0
CIDR notation	/24
Usable host range	192.168.10.1 – 192.168.10.254
Broadcast address	192.168.10.255
Total usable IP addresses	254

2. IP Address Allocation

Device name	IP address	Subnet mask
Computer 1	192.168.10.2	255.255.255.0
Computer 2	192.168.10.3	255.255.255.0
Computer 3	192.168.10.4	255.255.255.0
Computer 4	192.168.10.5	255.255.255.0
Computer 5	192.168.10.6	255.255.255.0
Network Printer	192.168.10.10	255.255.255.0

3. Addresses Reserved for Future Expansion

Reserved IP range	Purpose
192.168.10.11 – 192.168.10.50	Future computers and printers
192.168.10.51 – 192.168.10.100	Servers and other network devices
192.168.10.101 – 192.168.10.254	Additional future devices

These addresses are reserved for future use and should not be assigned to other devices until required.

4. Simple Explanation

- Network address: Identifies the network.
- Subnet mask: Identifies the network and host portions of an IP address.
- Usable host range: The IP addresses that can be assigned to devices.
- Broadcast address: Sends traffic to all devices on the subnet.
- Reserved addresses: Keep IP addresses available for future devices.

A `/24` subnet supports up to 254 usable IPv4 addresses, which is sufficient for the six current devices and future expansion.

5. Result

The IP addressing plan successfully allocates unique IP addresses to five computers and one network printer. It also identifies the subnet details and reserves addresses for future expansion.

STEP 2 · TEST & DEBUG

Then, working without AI, verify the plan for errors — overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself.

SUBMIT :1 The exact prompt(s) you gave the AI tool. 2 The AI's original subnetting plan and your corrected version. 3 A 3–4 line note explaining what you changed and why the AI's version needed it

Exact Prompt Given to AI

Generate an IP addressing and subnetting plan for Nexus Innovations' startup office network with 5 computers and 1 network printer. Use a private Class C IPv4 subnet. Include the network address, subnet mask, usable host range, broadcast address, IP address for each computer and printer, and addresses reserved for future expansion.

2. AI's Original Subnetting Plan

Network detail	Original plan
Network address	192.168.10.0
Subnet mask	255.255.255.0
CIDR	/24
Usable host range	192.168.10.1 – 192.168.10.254
Broadcast address	192.168.10.255

Device	IP address
Computer 1	192.168.10.2
Computer 2	192.168.10.3
Computer 3	192.168.10.4
Computer 4	192.168.10.5
Computer 5	192.168.10.6
Network Printer	192.168.10.10

Original reserved ranges:

- 192.168.10.11 – 192.168.10.50: Computers and printers.
- 192.168.10.51 – 192.168.10.100: Servers and other devices.

3. My Corrected Subnetting Plan

After checking the original plan, the network and broadcast addresses are correct. The subnet has enough host capacity. I have clarified the reserved ranges to avoid overlap with currently assigned addresses.

Network detail	Corrected plan
Network address	192.168.10.0
Subnet mask	255.255.255.0
CIDR	/24
Usable host range	192.168.10.1 – 192.168.10.254
Broadcast address	192.168.10.255
Total usable hosts	254

Corrected device IP allocation

Device	IP address
Computer 1	192.168.10.2
Computer 2	192.168.10.3
Computer 3	192.168.10.4
Computer 4	192.168.10.5
Computer 5	192.168.10.6
Network Printer	192.168.10.10

Corrected reserved ranges

IP range	Purpose
192.168.10.11 – 192.168.10.50	Future computers and printers
192.168.10.51 – 192.168.10.100	Future servers and network devices
192.168.10.101 – 192.168.10.254	Additional future expansion

These reserved ranges do not overlap with the six assigned device addresses.

4. My 3–4 Line Note

I checked the AI-generated subnetting plan for network, broadcast, and host address errors. The network address, subnet mask, broadcast address, and device IP addresses were correct. I clarified the reserved IP ranges to avoid overlap with assigned devices and provide space for future expansion. The /24 subnet provides 254 usable IP addresses, which is sufficient for the office network.The subnetting plan was verified and the reserved ranges were clearly organized for future growth.